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Assessing the Effects of Macroalgae Beds on Coastal Sedimentary Nematodes in the South Caspian Sea

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Abstract
In this study, the null hypothesis is that sediment and overlying water parameters, biological and functional diversities of benthic nematodes, do not differ between macroalgal beds and adjacent unvegetated sediments. Samples were collected from unvegetated sediments and adjacent sediments beneath the macroalgae beds at three stations along the south coast of the Caspian Sea. Two species of macroalgae, *Cladophora glome*rata and *Ulva intestinalis*, were identified. Sediments under the macroalgal bed had a higher content of microplastics and total organic matter as well as a higher dissolved oxygen compared to unvegetated sediments. A total of 16 nematode species belonging to 14 genera were identified, with *Daptonema*, represented by three species, being the numerically dominant genus. Both total and vertical nematode densities, as well as diversity indices, were higher in vegetated sediments at each station. Analysis of community structure revealed notable differences between the two habitats, due mostly to the presence or absence of certain nematode species. These variations in nematode community characteristics may be linked to the higher dissolved oxygen levels in the overlying water and the increased total organic matter and food in vegetated areas. Macroalgae in coastal areas can serve as biotic retention zones for microplastics, enhance nematode densities and diversity, and alter functional diversity and community structure. The retreat of the Caspian Sea will lead to the loss of this habitat, which may result in declines of several meio- and macrofauna taxa closely associated with macroalgal beds.

Keywords Macroalgae · Free-living nematode · Biodiversity · Microplastics · Ecosystem engineer · The caspian sea

Introduction

Macroalgae play an important role in coastal ecosystems by providing food and habitat for various marine organisms, such as small invertebrates and fishes. Macroalgal beds can stabilize the sediment, enhance organic matter accumulation, increase the food availability, and protect physically against waves and water disturbance (Baden et al., 2010; Armenteros et al., 2020). They present all along the coast and may cover hard substrates or soft sediments and serve as sheltered habitats with high productivity for aquatic animals hence, they can be regarded as ecosystem engineers that alter the sedimentary environment (Fredericq et al., 2019; Bazterrica et al., 2021) for benthic organisms such as free-living nematode (meiofauna) communities in sediments (Liao et al., 2015; Ruiz-Abierno & Armenteros, 2017; Gallucci et al., 2019).

The Caspian Sea is the largest landlocked non-tidal brackish water lake in the world. Between 1977 and 1995, the sea level of the Caspian Sea rose by ~4 m, resulting in flooding of inland areas (Jafari, 2010). This led to significant erosion of coastal regions and their infrastructure. In order to address this issue, both natural and artificial rock walls (boulders) were constructed along the southern coast to prevent further flooding of inland areas. Coastal rocks provide an ideal habitat for macroalgae due to their hard

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surfaces and stable substrate for attachment, and macroalgae are commonly found attached to rocks, boulders, and other substrates including sand in this area. In total three species of macroalgae, two green (*Cladophora glomerata* and *Ulva intestinalis*) and one red (*Osmundea caspica*) have been identified (Mehdipour et al., 2015). Unlike the South Caspian, the northern part of the sea is heavily influenced by the Volga River and is characterized by low salinity, shallow waters, silty soils, and low transparency, which consist of macroalgae and flowering plants (Blinova, 2007; Stepanian, 2016). One specific effect of global warming in the Caspian Sea is shrinking of the sea surface area (Prange et al., 2020; Samant & Prange, 2023) as a result of increasing surface water temperature (~3.2 °C) over 35 years (Beyraghdar Kashkooli et al., 2019), increasing the rate of evaporation and changes in the annual cycle of wind directions and the amount of water flowing into the Caspian Sea from the Volga River, which is the largest river flowing into the lake (Azizpour & Ghaffari, 2023). The retreat of the Caspian Sea has significant ecological impacts on coastal ecosystems including loss of habitat, especially for macroalgal communities, changes in water chemistry, increase in salinity, especially in bays and lagoons, and coastal erosion (Jamshidi, 2017; Lahijani et al., 2023). Currently, the sea level has decreased rapidly and many of the artificial stone structures are no longer underwater, leaving only a few areas with natural stones and macroalgal beds. As macroalgal beds play an important structuring role in coastal areas (e.g. Fonseca et al., 2011; Gallucci et al., 2019), their loss on the south Caspian Sea coasts can have negative ecological consequences for the composition and dynamics of the associated benthos (Ghasemi et al., 2013). In present study, we tested the null hypothesis (H0) that the environmental conditions (sediment and overlaying water parameters) and the biological (density, diversity indices, vertical distribution, and community structure) and functional (Maturity Index and Trophic Diversity Index) diversities of free-living nematodes did not differ between macroalgal beds and adjacent unvegetated sediments across three different stations.

Materials and Methods

Field Sampling and Experimental Procedures

Sampling was carried out in September 2023 on the southern coasts of the Caspian Sea (Fig. 1). Three stations, from east to west—Sisangan, Chalus, and Ramsar—whose rocky coasts are still underwater after the recent retreat of the Caspian Sea, were investigated (Table 1). First, at each station, attached macroalgae on hard substrata were collected by scraping a surface area of 0.04 m^2 in triplicate. The macroalgae were washed gently with fresh water and all attached fauna and sediments were removed. Species were identified based on gross morphology and internal features (Burrows, 1991; Leliaert & Coppejans, 2003). Total and each species biomass were determined as dry weight after oven-drying at 60 °C for 24 h (Aguila Ramirez et al., 2003) and represented as biomass

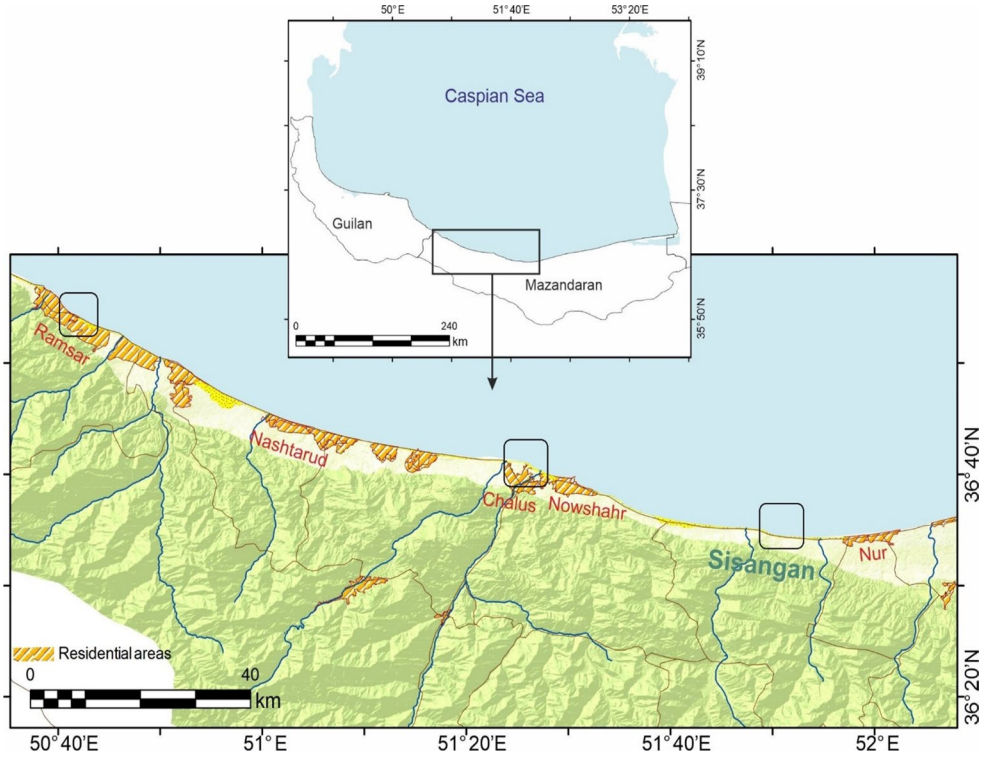

Fig. 1 Sampling stations from the coastline of the south Caspian Sea in Mazandaran province. From east to west: Sisangan, Chalus and Ramsar

Table 1 Environmental parameters measured in sampling stations. Different lower case letters indicate differences (pairwise test) between habitats in each station

	Sisangan vegetated	Sisangan unvegetated	Chalus vegetated	Chalus unvegetated	Ramsar vegetated	Ramsar unvegetated
MP_S (number)	1.7 ± 0.6^a	0.3 ± 0.6^b	1.0 ± 1.0^a	0.7 ± 0.6^b	2.7 ± 0.6^a	0.7 ± 0.6^b
DO (mg/L)	9.4 ± 0.5^a	7.8 ± 0.2^b	8.9 ± 0.6^a	6.3 ± 0.3^b	9.8 ± 0.2^a	6.5 ± 0.2^b
pH	8.1 ± 0.0^a	7.9 ± 0.2^a	7.8 ± 0.0^a	7.9 ± 0.1^a	7.7 ± 0.3^a	7.8 ± 0.2^a
Gravel (%)	19.1 ± 2.1^a	15.2 ± 0.9^b	13.1 ± 0.1^a	12.9 ± 0.1^a	3.5 ± 0.1^a	2.4 ± 1.6^b
Sand (%)	68.7 ± 1.1^b	71.4 ± 2.2^a	77.1 ± 0.6^a	77.4 ± 0.4^a	85.8 ± 0.2^a	85.3 ± 0.2^a
Silt + Clay (%)	12.2 ± 2.0^a	13.3 ± 0.7^a	9.7 ± 0.6^a	9.2 ± 0.4^a	10.6 ± 0.1^a	12.2 ± 0.2^a
TOM (%)	5.7 ± 0.2^a	3.9 ± 0.4^b	3.5 ± 0.1^a	1.5 ± 0.4^b	9.0 ± 0.5^a	6.2 ± 0.5^b
Phosphate (μg/L)	37.0 ± 4.6^a	48.7 ± 8.1^a	12.0 ± 2.6^a	15.3 ± 1.5^a	40.3 ± 14.0^a	50.0 ± 15.7^a
Nitrate (μg/L)	201.3 ± 13.0^a	203.3 ± 7.6^a	123.3 ± 13.0^a	109.7 ± 11.9^a	291.7 ± 0.6^a	262.7 ± 0.6^a

per 1 m^2. All sediment samples were taken at least 5 m apart at 0.5-m water depth on a line parallel to the beach (Taheri et al., 2024). Simultaneously at each station, three sediment samples in unvegetated sediments and sediments underneath the macroalgal patch were sampled using a hand corer (i.d. 4 cm, surface 12.56 cm^2). For each replicate, a corer was inserted to 5-cm depth for nematode studies. In the field, sediments were sliced at 1-cm intervals down to 5 cm depth and stored in clean plastic containers, fixed in 4% buffered formaldehyde solution and stained with Rose Bengal. In the laboratory, each sample slice was sieved over 1000- úm and 38- úm sieves. The fraction remaining on the 38-úm sieve was centrifuged three times with Ludox (specific gravity 1.18) to separate organisms from sediments and then counted under a stereomicroscope. From every slice, 120 nematodes (or all nematodes if a lower number was observed) were hand-picked (Taheri et al., 2014a, 2015) and were transferred to pure glycerin via a dehydration procedure (Vincx, 1996) and mounted on microscopic slides for identification to species level using the NeMys online identification system (Nemys, 2023). The identified nematodes were assigned to different feeding types based on the morphology of the buccal cavity: (1 A) selective deposit feeders; (1B) non-selective deposit feeders; (2A) epistrate (diatom) feeders; (2B) predators/omnivores (Wieser, 1953). Accordingly, the index of trophic diversity (ITD), which is based on the composition of nematode trophic guilds reflected by the buccal cavity shape and the presence/absence of cuticularization, was calculated following Heip et al. (1985). ITD values range from 0.25 (highest trophic diversity) to 1.0 (lowest trophic diversity). The Maturity index (MI) is based on the life and reproductive strategies of free-living nematodes ranging from extreme r-strategist species (colonizers or c-p 1) and k-strategist species (persisters or c-p 5). This index (Bongers, 1990; Bongers et al., 1991) was calculated as the weighted average of the individual colonizer-persistent (c-p) values.

Another three replicate sediment samples were taken at each station in unvegetated and vegetated sediments using a hand corer (i.d. 4 cm, surface 12.56 cm^2) to measure the % total organic matter (TOM) and sediment grain size. Sediment from the surface (2 cm) was subsampled and stored in a cleaned plastic container. Total organic matter (TOM) was determined by weight loss on ignition (4 h at 550 °C) after drying (24 h at 90 °C) to a constant weight (Heiri et al., 2001). Grain size distribution of the sediment was determined using a Particle Size Analyzer (HORIBA-LA950, France & Japan) and the sediment fractions (gravel, sand, silt + clay) were reported as percentages and defined based on the size of particles in the Wentworth scale (Wentworth, 1922). At each station, simultaneously with sediment sampling, overlaying water dissolved oxygen and pH were measured using a portable multimeter (HACH HQ40d, USA) in unvegetated and vegetated areas. Separately, 100 mL of overlaying water were collected to measure concentrations of nitrate and phosphate with a spectrophotometer (Analytik-jena, SPECORD-210) using standard analytical protocol (Murphy & Riley, 1962; APHA, 2005).

At each station, three sediment samples from the top 5 cm were collected using hand corer (i.d 10 cm) in both unvegetated and vegetated areas. The samples were placed in clean Mason jars covered with aluminum foil and dried at 50 °C for 72 h (Wang et al., 2017). Subsequently, 800 g of dry sediment from each replicate was evenly distributed (200 g each) into four 2000-mL glass beakers, and microplastics were isolated by adding 1500-mL of zinc chloride separation solution with a specific density of 1.55 g/mL to each beaker. The sediments were then stirred for 10 min with a clean stainless steel spoon, and after 1 h of settling, the supernatant was filtered using a 38-úm mesh sieve. Particles on the sieve surface were washed twice with distilled water into a clean 1000-mL Mason jar (Yazdani Foshtomi et al., 2019). Subsequently, 25-mL of 35% hydrogen peroxide were added to each 1000-mL Mason jar to eliminate natural organic matter. The jars were covered with aluminum foil and left overnight. To dissolve shell fragments, 25-mL of 4.5% HCl solution were introduced to each jar. The contents of each jar were then filtered on filter paper, which was rolled up, secured, and placed in clean Petri dishes. These were air-dried at room temperature for 24 h (Zobkov & Esiukova, 2017). All microplastic particles (MP_S) were examined under a Leica Wild M8 stereomicroscope at magnifications up to 40× and counted.

Statistical Analysis

Total number of species, Shannon diversity (H', log e), and evenness (Pilou's, J) were calculated for all stations using PRIMER v6 with PERMANOVA add-on (Anderson et al., 2008). To find different groups of sampling stations based on the environmental variables, principal components analysis (PCA) was used (Clarke & Warwick, 1994). Possible differences in the biomass of macroalgae (per g dw/m^2, univariate) in different stations (St) were investigated using a one way PERMANOVA.

Possible differences in total nematode density, total number of species, Shannon diversity, evenness, MI and ITD as well as all measured environmental variables two–way PERMANOVA (station*habitats, both fixed factors) was used. Possible differences in the vertical profiles of nematode density, total number of species, Shannon diversity and evenness (univariate) and community structure (multivariate) among stations were investigated using a fully crossed three factor design using PERMANOVA. The design included the random factor replicate (Rep) nested in the fixed factor stations and the fixed factor habitat. In all mentioned tests, whenever significant differences were observed, pairwise tests (p-values were obtained using a Monte Carlo test) were performed to investigate differences (Anderson & Robinson, 2003). A non–metric Multidimensional scaling plot (nMDS) based on Bray–Curtis similarity visualized the vertical nematode community structure. Furthermore, one–way SIMPER analysis based on nematode abundance of the full sediment column (0–5 cm) was used to identify the nematode species making important contributions to within-group similarity. The cut-off for a low contribution was 100% (Clarke & Warwick, 1994). All above analyses were done in PRIMER v6 with PERMANOVA + add-on.

Results

Biomass of Macroalgae

Two species of macroalgae, *Cladophora glomerata* and *Ulva intestinalis* were observed at all stations. The species, *C. glomerata* with more than 90% of total biomass was dominant in all three stations. The total biomass of macroalgae was different among stations (Pseudo-F = 50.53, $p = 0.001$, $n = 3$). Biomass was higher at the Ramsar station (398 ± 13, g dw/m^2) followed by the Sisangan station (256 ± 13, g dw/m^2) and the Chalus (82 ± 11, g dw/m^2) station.

Environmental Conditions

Environmental conditions are listed in Table 1. Total number of microplastics in cores differed among habitats (Pseudo-F = 15.12, $p = 0.005$, $n = 3$). In all stations the highest number of MP_S (1–2.66) was counted in vegetated areas (t = 3.89, $p = 0.001$, $n = 3$). Results of the two–way PERMANOVA for dissolved oxygen (DO) in water (mg/L) showed differences between habitats and stations. DO was higher (8.96–9.77) at vegetated areas in comparison with unvegetated areas in all stations. There were no differences in pH between habitats and stations. There were slight differences in sediment granulometry (% gravel, sand, and silt-clay) between habitats and stations. The total organic matter values (%) varied between habitats and stations. TOM was higher (3.51–8.99) in vegetated areas than unvegetated areas at all stations. There were no differences in phosphate and nitrate values (µg/L) between vegetated and unvegetated areas (Table 1).

The result of the PCA is shown in Fig. 2. PCA1 and PCA2 accounted for 64.00% and 19.90% of the total variance, respectively.

Nematode Community Structure

In the present study, 16 species belonging to 14 genera were identified. The genus *Daptonema* with three species was the numerically dominant group. Total nematode density was statistically different among stations and habitats (Pseudo-F = 20.17, $p = 0.001$, $n = 3$). Total nematode density was higher (two and four times) in vegetated than unvegetated

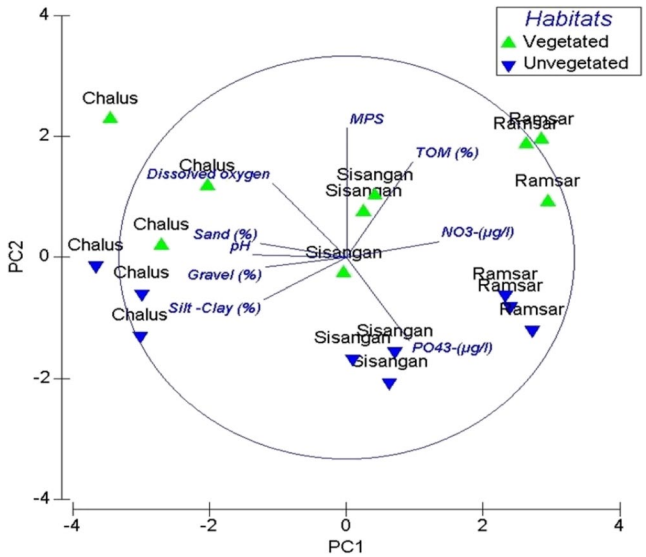

Fig. 2 Principal component analysis based on environmental data

areas. The number of species differed among stations and habitats (Pseudo-F = 26.45, $p = 0.001$, $n = 3$). There were no differences in total number of species at Sisangan station while the higher number were counted in vegetated areas at Chalus and Ramsar stations. The evenness index was different among stations and habitats (Pseudo-F = 13.57, $p = 0.004$, $n = 3$). The evenness index was different between habitats and was higher at Sisangan and at Chalus stations, respectively. Shannon diversity was different among stations and habitats (Pseudo-F = 10.55, $p = 0.001$, $n = 3$). This index did not differ at Sisangan station while their values were higher at vegetated areas at other stations (Fig. 3).

The Maturity Index (MI) and the index of trophic diversity (ITD) differed between habitats (Pseudo-F = 37.26, $p = 0.001$, $n = 3$ and Pseudo-F = 206.43, $p = 0.001$, $n = 3$, respectively). The highest values of MI were obtained in vegetated areas at Sisangan and Ramsar stations. The higher values of ITD were obtained in the vegetated area at Sisangan and in unvegetated areas at Chalus and Ramsar, respectively (Supplementary Information Fig. S1).

The SIMPER analysis showed that the relative abundances of species and their feeding types differed between nematode communities in vegetated and unvegetated sediments. This was most at Sisangan, where the proportion of non-selective deposit feeders (1B) was 77.7% in unvegetated sediment, contrasting with 4.0% in vegetated sediment. At this station, the percentage of predators/omnivores species (2B) increased approximately six fold (49.1%) in vegetated sediment compared to unvegetated sediment (8.1%). Additionally, the relative abundance of epistrate (diatom) feeders (2 A) was 13.2% in unvegetated sediment to 46.9% in vegetated sediment. At the Chalus station, the community in unvegetated sediment was entirely dominated by predators/omnivores species (2B), whereas three species of epistrate (diatom) feeders (2 A) were observed in vegetated sediment, accounting for 19.0% of the total community. At the Ramsar station, the community in unvegetated sediment was completely dominated by non-selective deposit feeder species (1B), while predators/omnivores species (2B) accounted for 11.1% of the entire community in vegetated sediment (Table 2).

The vertical distribution of the nematode density was affected by the interaction (Pseudo-F = 6.77, $p = 0.001$, $n = 3$). In general, highest densities were observed in the upper 3 cm of sediments in comparison with deeper layers (4- and 5-cm depths). With the exception of the 3 cm layer at Sisangan, densities in vegetated areas were higher than unvegetated areas in the upper 3 cm. There were no differences between habitats in the deeper layers (4- and 5-cm depths). The vertical distribution patterns of the number of species was affected by the interaction (Pseudo-F = 8.02, $p = 0.002$, $n = 3$). There were no differences between habitats at Sisangan station while differences at the other two stations showed no trends. Evenness indices were considerably different among the interaction terms (Pseudo-F = 16.80, $p = 0.001$, $n = 3$). In general, there was difference in evenness index at each station between habitats. Shannon diversity was different among the interaction terms (Pseudo-F = 16.07, $p = 0.001$, $n = 3$). At Sisangan station in upper three cm, higher values of Shannon index were obtained in unvegetated areas while there were no differences between habitats in the deeper layers (4 and 5 cm depths). In general, at the other two stations the trends were reversed (Supplementary Information Fig.

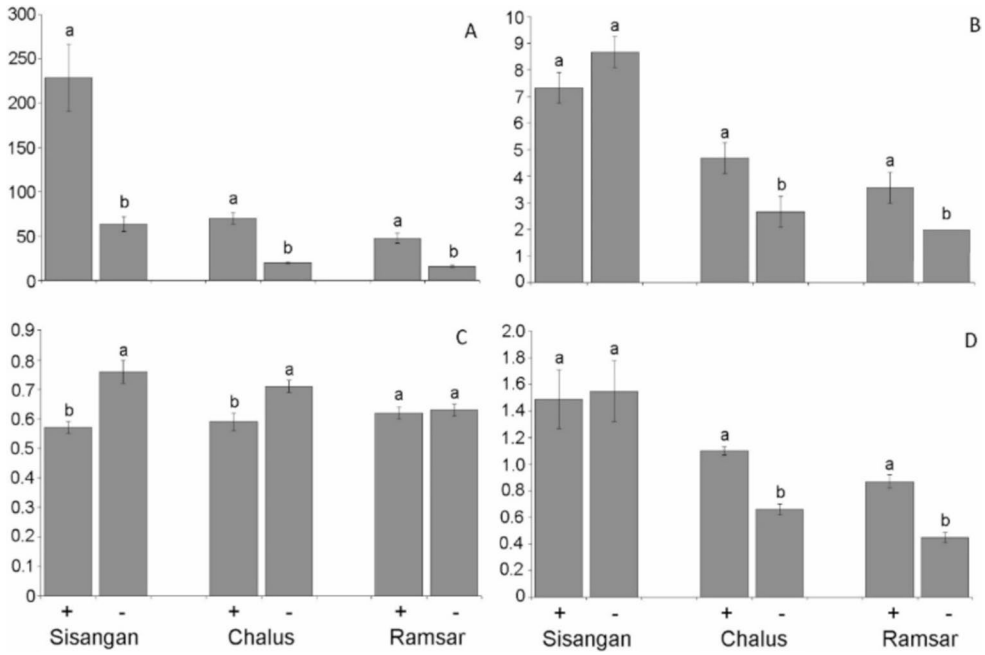

Fig. 3 Total nematode densities (**A**), number of species (**B**), evenness (**C**) and Shannon diversity (**D**) in different stations. Different lower case letters indicate differences (pairwise test) between habitats at each station. "+"= vegetated sediment and "-" = unvegetated sediment

Table 2 Results of one–way SIMPER analysis for nematode communities and feeding types per station. FT = feeding type, Cont = contribution. (1 A) selective deposit feeders; (1B) non-selective deposit feeders; (2 A) epistrate (diatom) feeders; (2B) predators/omnivores

Sisangan vegetated			Sisangan unvegetated		
Average similarity: 65.6	Cont	FT	Average similarity: 66.5	Cont	FT
Adoncholaimus aralensis	47.0	2B	*Daptonema karabugasensis*	39.5	1B
Chromadorella parapoecilosoma	33.8	2 A	*Daptonema intermedia*	20.7	1B
Chromadorissa beklemishevi	7.9	2 A	*Daptonema curticauda*	17.9	1B
Chromadorina germanica	5.8	2 A	*Adoncholaimus aralensis*	7.8	2B
Anoplostoma viviparum	3.5	1B	*Oncholaimus hyrcanus*	5.4	2B
Oncholaimus hyrcanus	2.1	2B	*Chromadorella parapoecilosoma*	3.9	2 A
Axonolaimus spinosus	0.2	1B	*Neochromadora poecilosoma*	3.1	2 A
Tripyloides marinus	0.2	1B	*Chromadorita tenuis*	0.7	2 A
			Microlaimus naidinae	0.4	2 A
Chalus vegetated			**Chalus unvegetated**		
Average similarity: 47.3	Cont	FT	Average similarity: 65.1	Cont	FT
Halodorylaimus micramphis	51.9	2B	*Halodorylaimus micramphis*	85.5	2B
Adoncholaimus aralensis	29.1	2B	*Adoncholaimus aralensis*	7.5	2B
Chromadorella parapoecilosoma	14.1	2 A	*Enoploides fluviatilis*	7.0	2B
Chromadorissa beklemishevi	3.4	2 A			
Chromadorina germanica	1.5	2 A			
Ramsar vegetated			**Ramsar unvegetated**		
Average similarity: 74.6	Cont	FT	Average similarity: 83.5	Cont	FT
Daptonema karabugasensis	67.3	1B	*Daptonema karabugasensis*	86.6	1B
Axonolaimus spinosus	21.5	1B	*Axonolaimus spinosus*	13.4	1B
Halodorylaimus micramphis	9.5	2B			
Adoncholaimus aralensis	1.7	2B			

S2). Nematode community structure was not affected by the interaction term (Pseudo-F = 109.49, $p = 0.001$, $n = 3$). The nMDS plot (Fig. 4) shows a difference in nematode community structures between habitats.

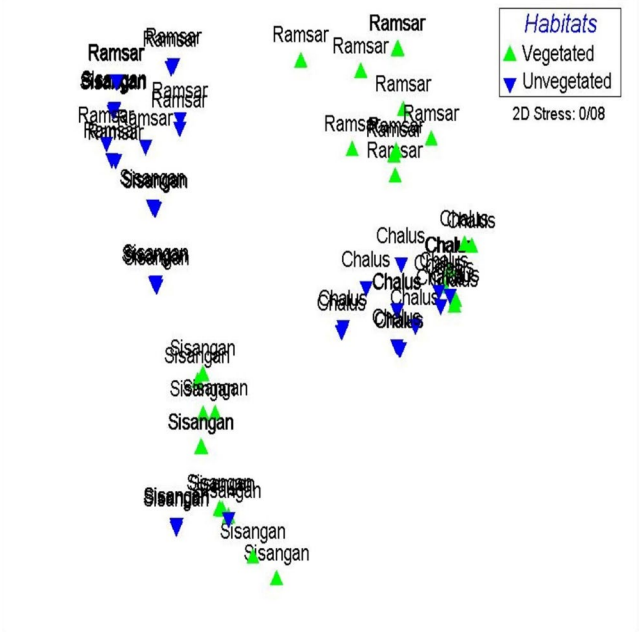

Fig. 4 nMDS plot based on Bray-Curtis similarity on nematode vertical density data

Discussion

In the present study two species of macroalgae, *Cladophora glomerata* and *Ulva intestinalis* were identified. There was a difference in total biomass of macroalgae among stations with a higher value at Ramsar, followed by Chalus and Sisangan. The species, *C. glomerata* with more than 90% of total biomass was dominant. Similar species have been reported in previous studies in the south Caspian Sea (Mehdipour et al., 2015). Our results showed that the macroalgal bed clearly changed the sedimentary environment and had a significant role in increasing the density, diversity and shaping community structure of free-living nematodes when compared to unvegetated habitats at each station. Therefore, we reject the null hypotheses (H0).

Our findings indicate the presence of macroalgae in coastal ecosystems was linked to a notable increase in the quantity of microplastics in surface sediments of vegetated areas, aligning with recent research (e.g. Li et al., 2022; Zhang et al., 2022). *Cladophora glomerata* is a branched filamentous alga, so debris can get caught in the fronds, resulting in the buildup of plastic within the sediment bed (Li et al., 2022). The accumulation of microplastics in macroalgae beds has been noted in a number of studies (Peller et al., 2021). Research in the Baltic Sea has shown that filamentous algae accumulate more microfibers than other algae. Research on the accumulation of plastic debris in

macroalgae beds in the southern Caspian Sea is limited, but there is growing concern about the potential effects of plastic pollution on these vital aquatic ecosystems (Manbohi et al., 2021, 2023).

Decomposition of dead macroalgae can lead to an increase in sedimentary organic matter. We found a higher percentage of sedimentary organic matter under the vegetated habitat compared with unvegetated sediments in agreement with previous studies (e.g., Fonseca et al., 1983, 2011; Fonseca & Bell, 1998; Baden et al., 2010; Zhang et al., 2022). At all stations, water underneath the macroalgal bed was more oxygenated than water over unvegetated sediments while there was no difference in sedimentary texture, which contrasts with previous study (e.g. Fonseca et al., 2011). Macroalgae are typically considered ecosystem engineers that reduce waves and water disturbance (physical stress), stabilize the sediment, increase organic matter accumulation, and protect smaller invertebrates from predators (e.g. Orth et al., 1984; Baden et al., 2010; Fonseca et al., 2011; Armenteros et al., 2020; Berezina et al., 2022).

The Caspian Sea exhibits low biodiversity, with a significant portion of its fauna being endemic to the region (Dumont, 2000). Biodiversity is 2.5 times lower than in the Black Sea and 5 times lower than in the Barents Sea. Its brackish water has too high a salinity for true freshwater species but too low for species of marine origin. These conditions are favorable only for brackish water species (Mordukhai-Boltovskoi, 1979); Karpinsky, 2005). So far, 50 species of free-living nematodes have been reported in the whole Caspian Sea from fresh and brackish waters (Tchesunov, 1980, 1981a; 1985; Mokievsky & Miljutina, 2011). In the present study 16 species belonging to 14 genera were identified and the genus *Daptonema* with three species was the numerically dominant group which is in agreement with previous studies (Taheri et al., 2014b, 2017; Darvish Bastami et al., 2017). One species of freshwater origin (*Halodorylaimus micramphis*) was observed at the Chalus and Ramsar stations, where some freshwater streams flow to the coast. This species was previously reported by Tchesunov, 1980, 1981ab; 1985.

Higher total and vertically separated nematode densities (between two and four times) were observed at all stations in vegetated areas. This was more prominent at Sisangan, where density was about four times higher in vegetated sediment. Whole, the total number of species and diversity indices also differed between the two habitats; in vertical distribution the highest values were observed in the upper 3 cm of the sediments in comparison with the deeper layers. Two explanations for these higher densities are possible. First, it has been demonstrated that macroalgae (*Cladophora*) can create a habitat for microbial (Wang et al., 2023) and microalgal communities (Prazukin et al., 2020) which in turn can provide many ecological niches fostering meiofauna and free-living nematodes, which may feed on these microorganisms (Somerfield et al., 2002; Berezina et al., 2022) and as a result increase their density. Moreover, macroalgae may provide food for some species that feed directly on macroalgal tissues (Frame et al., 2007; Giere, 2009). Second, the waters of the south Caspian Sea are continuously moving because of wave action and rip currents (Dehbashi & Azarmsa, 2021; Mahmoudof & Mohammadali, 2022). Macroalgal beds may protect and stabilize the sediments from wave and rip current backwash exposure, which in turn structure meiofaunal and free-living nematode communities (McLachlan & Hesp, 1984; Giere, 2009). Higher density and diversity indices and different community structure have been reported under macroalgal/seagrass beds in comparison with bare unvegetated sediment (e.g. Liao et al., 2015, 2016; Rizqydiani et al., 2018; Berezina et al., 2022; Mohammad, 2022).

Relative abundance (which can explain differences in total densities) of observed species and their feeding type contribution (functional aspect) also changed between nematode communities in vegetated and unvegetated sediments. This change was most prominent at Sisangan, where the relative abundance of non-selective deposit feeders (1B) was 77.7% in unvegetated sediment while it was 4.0% in vegetated sediment. In this station, relative abundance of predators/omnivores species (2B) increased about 6 times (49.1%) in vegetated sediment in comparison with unvegetated sediment (8.1%). The relative abundance of epistrate (diatom) feeders (2 A) increased from 13.2% in unvegetated sediment to 46.9% in vegetated sediment. At Chalus station, the community was dominated (100%) by predators/omnivores (2B) species in unvegetated sediment while three species of epistrate (diatom) feeders (2 A) were observed in vegetated sediment, comprising 19.0% of the whole community. Finally, at Ramsar, the unvegetated sediment community was dominated (100%) by non-selective deposit feeder (1B) while in vegetated sediment predators/omnivores species (2B) made up 11.2% of the community. The differences in density, community structure and functional diversity between vegetated and unvegetated habitats in the current study confirms the role of the macroalgal bed as an ecosystem engineer in the south Caspian Sea (Wright & Jones, 2006; Fonseca et al., 2011; Liao et al., 2015; Ruiz-Abierno & Armenteros, 2017). At present, almost all marine ecosystems are under pressure from microplastic pollution (e.g. Kershaw, 2015), and macroalgae beds are zones of biotic retention of this component.

Our study shows that macroalgal bed alter sedimentary environments and serve as natural shelters for various nematode species. The habitat modifications created by these macroalgal beds appear to enhance local density

and biodiversity of free-living nematodes, although this effect varies from place to place. At all stations, the sediments beneath the macroalgae beds has higher levels of total microplastics, organic matter, and dissolved oxygen compared to unvegetated sediments. Given the presence of macroalgae in both natural and artificial rock formations (boulders) in shallower water, the retreat of the Caspian Sea could have detrimental ecological impacts on coastal ecosystems, including the loss of macroalgal habitats. The disappearance of these habitats may lead to the decline of various free-living nematodes that are closely associated with macroalgal beds, potentially resulting in shifts within nematode functional groups.

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Data Availability The datasets in the current study are available from the corresponding author on reasonable request.
Competing Interest.
The authors declare that they have no known competing financial interests or personal relationships that could appear to influence the work reported in this manuscript.

Declarations

Ethical Approval and consent to participate No specific permits for sampling and ethics requirements were needed since our research was linked on intertidal marine nematodes.

Consent for Publication Not applicable.

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